

Basic Machine Learning II

Generalisation, evaluation, and model selection

Jonathon Hare

Training performance is not the final objective

Underfitting and overfitting

Capacity, data, and inductive bias

Regularisation changes the learning problem

Data augmentation encodes invariances

Metrics must match the task

Thresholds and class imbalance

Validation supports model selection

Data leakage invalidates evaluation

Reporting uncertainty and variation

A defensible evaluation protocol